A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 32

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1** Spoil contains residual sulfide ores (eg iron pyrites, zinc sulfide) **1** These react with oxygen in the air spaces between the waste particles (oxidation) **1** Producing metal oxides and oxides of sulfur which dissolve in drainage water to form sulfuric acid **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** The low pH (3.4) directly damages exposed cells / denatures proteins and dissolves calcium-based exoskeletons **1** Low pH increases the solubility of zinc (0.02 to 6.5 mg dm^{-3}) which is toxic (inhibits enzymes) **1** Only a few tolerant taxa survive – the sensitive mayfly larvae (indicator of clean water) are eliminated – dissolved oxygen is not the limiting factor here **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **limiting factor** → the factor in shortest supply that holds the rate or population back	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Dilution by tributaries / more water **1** Neutralisation by alkaline rocks (limestone) in the stream bed – zinc precipitates / adsorbs onto sediments as pH rises **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks.	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Collect the drainage water and pass it through crushed limestone / add lime to neutralise the acid **1** Higher pH also reduces the solubility of toxic metals so they precipitate and can be removed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 273-275; AQA spec 3.4.3.2.9-3.4.3.2.10.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**1 mark each** 1 mark for each of the 4 points below. **1** Surplus electricity (eg windy nights) electrolyses water to produce hydrogen which is stored **1** Hydrogen has a high energy density so it can be transported / stored for long periods **1** Released by combustion for heat, industrial processes or steam turbines **OR** by fuel cells combining hydrogen and oxygen electrochemically to produce electricity and water **1** Can also be used as a vehicle fuel or converted to methane / ammonia **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	4
02.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Compressed gas (up to 700 times normal volume) - energy to run the compressor, strong tanks **1** Liquefied at very low temperature - energy for refrigeration and compression **1** Adsorbed on a metal matrix (metal hydride) at low pressure - tanks larger and heavier than a petrol tank storing the same energy **1** Conversion to ammonia - some energy needed but less than the alternatives **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Both convert chemical energy to electricity electrochemically, but a battery stores a fixed amount of chemicals which are reformed by recharging **1** A fuel cell keeps producing electricity as long as fresh fuel (hydrogen / alcohol) and oxidant are supplied **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **fuel cell** → a cell that produces electricity from hydrogen and oxygen	2

Do not write outside the box

Question	Marking guidance	Marks
02.4	1 mark for the point below. **1** Using hydrogen made from surplus renewable electricity as the main store and carrier of energy, released when demand exceeds supply or intermittent sources fail, for heating, vehicles and electricity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 219-221; AQA spec 3.3.4.1.	
03.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 C:** Organic (methyl) mercury **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	1
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Anaerobic microbes converted the inorganic mercury into organic methyl mercury **1** Methyl mercury is liposoluble so it is absorbed, crosses into the brain and biomagnifies along the food chain to fish eaten by people **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Most heavy metals are less soluble at high pH **1** So they remain immobile and are not leached into water or absorbed by organisms **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 6 points below. **1 Direct effect of abiotic change:** species near the edge of their range of tolerance (temperature, water) may die out locally, eg sundews if marshes dry, frogs if climate drier – beech (shallow roots) suffers in drought more than oak **1 Changes in timing:** earlier flowering / emergence may separate flowers from pollinators – plant toxins building up earlier may kill caterpillars – warmer winters disturb dormouse hibernation so they use fat reserves and starve **1 Biotic changes:** food supply, predators, disease vectors (mosquitoes / malaria spreading), pollinators and seed dispersers may change **1 Distribution changes:** species may colonise newly suitable areas (triton molluscs around the UK) but slow-growing species (trees) may disappear from old ranges faster than they colonise – suitable areas or biological corridors may not exist – human land use blocks movement **1 Population fragmentation:** rising seas or upward retreat to mountains isolate populations – divided gene pools increase inbreeding – small populations may not be viable – no repopulation from elsewhere **1** Some species benefit (bats: warmer winters, more insects) but wetter stormier weather reduces feeding time **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here. • **abiotic** → non-living / physical factors • **range of tolerance** → the range of conditions in which a species can survive • **Biotic** → living factors / effects of other organisms	6
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** All the species it depends on (food plants, pollinators, prey, mycorrhizae) will not move at the same speed **1** Less mobile species (plants, non-flying animals) cannot cross unsuitable areas without a biological corridor **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** The temperature difference between mid-latitude and polar air drives the jet streams – a smaller difference makes the jet stream slower and more meandering (Rossby waves) **1** Weather systems stall over an area, giving prolonged drought or rainfall / floods – unusually cold or warm air carried north or south **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 90-92; AQA spec 3.2.1.2.	
05.1	**1 mark each** 1 mark for each of the 3 points below. **1** A: Student's t-test - a difference between the means of two sets of normally distributed data **1** B: chi-squared test - a comparison of observed frequencies (counts) with expected frequencies **1** C: Spearman's rank correlation - a relationship between two variables measured on the same sampling points **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Student's t** → a test for a difference between the means of two normally distributed sets • **chi-squared** → a test comparing observed frequencies with expected frequencies • **Spearman** → a test for correlation between two sets of ranked data **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	3
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** There is a 2% probability, or a 1 in 50 chance, that a result at least as extreme as this would arise by chance alone if the null hypothesis were true **1** This is below the 5% significance level normally used, so the null hypothesis is rejected and the difference or correlation is described as statistically significant **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **null hypothesis** → a statement that there is no significant difference or correlation • **significance level** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance • **correlation** → a relationship in which the two variables change together **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it	2

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 3 points below. **1** The calculated value on its own has no meaning: it must be compared with the value that would be exceeded by chance in 5% of cases, which is read from a table of critical values **1** The critical value depends on the number of degrees of freedom, which comes from the number of categories or pairs of data, because a larger sample makes a given result less likely to be chance **1** If the calculated value exceeds the critical value at p = 0.05 the null hypothesis is rejected – if it does not, the null hypothesis is not rejected and the result is not significant **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **critical value** → the table value the calculated statistic must reach to be significant • **p = 0.05** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance • **null hypothesis** → a statement that there is no significant difference or correlation **Do not credit** • 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	3
05.4	**1 mark each** 1 mark for each of the 3 points below. **1** The standard deviation measures the spread of the individual values about the mean, so a small value means the heights were closely grouped and the mean is a good summary of them **1** Comparing the standard deviations with the difference between the means shows whether the two sets really are distinct or merely overlap **1** It is drawn as error bars above and below each plotted mean – where the error bars of two means do not overlap, a significant difference is likely, and where they overlap widely it is not **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean	3

Do not write outside the box

Question	Marking guidance	Marks
05.5	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** A correlation shows only that the two variables change together – it suggests a link that further work would have to test **1** A third factor may cause both, for instance shade, which raises soil moisture by reducing evaporation and also directly favours the moss **1** The relationship may run the other way round: a dense moss cover holds water and shades the soil, so the moss could be raising the moisture rather than responding to it **1** The points along one transect are close together and not independent, so a pattern may reflect one gradient across the site rather than a general relationship **1** Only an experiment in which soil moisture is changed while every other variable is held constant can establish cause **1** A result significant at p = 0.05 will still arise by chance about 1 time in 20, so a single significant result is not proof **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **correlation** → a relationship in which the two variables change together • **transect** → a line along which samples are taken to show change with distance • **p = 0.05** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • a correlation presented as proof of cause - say that it suggests a link that further work would have to test	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-407, 420-423; AQA spec 3.7.2.3.	
06.1	1 mark for the point below. **1** As temperature increases the maximum dissolved oxygen concentration decreases (non-linear: falls faster at low temperatures) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
06.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** About 39 °C (accept 38-40) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (°C) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1

Do not write outside the box

Question	Marking guidance	Marks
06.3	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 9.1 - 7.9 (value at 27 °C, accept 7.8-8.0) **1** = 1.2 mg dm^{-3} (accept 1.1-1.3) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (mg dm^{-3}) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** Figure 1 shows the maximum / saturation values – processes that remove oxygen (aerobic decomposition of organic matter / respiration) can give values below the line **1** Higher temperature increases the metabolic rate of organisms and the rate of bacterial decomposition so oxygen is used faster – values above the line are impossible **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.5	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Increased metabolic rate so more energy used / less surplus for growth **1** Eggs develop faster / hatch earlier (out of step with food supply) **1** Non-indigenous warm-water species may thrive and out-compete indigenous species **1** Toxicity of other pollutants is greater as toxins are metabolised faster **1** Reduced resistance to disease **1** Proteins / enzymes denatured if temperature is very high (above about 40 °C) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
06.6	1 mark for the point below. **1** Hot water is sprayed from a shower floor in a tall hollow tower – air absorbs the heat and rises, drawing in more air (updraught), so the water returned is cooler (and saturated with oxygen) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 256-257; AQA spec 3.4.3.2.6.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** C **Also accept** • **stratospheric ozone** → the ozone in the stratosphere that absorbs UV	1
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Stratospheric ozone is natural and beneficial: absorbs UV, protecting living organisms **1** Tropospheric ozone is a secondary pollutant (from NOx photochemistry) that is toxic to plants and humans and is a greenhouse gas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **Stratospheric ozone** → the ozone in the stratosphere that absorbs UV	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** HCFCs are less chemically stable / contain hydrogen so they are broken down in the troposphere **1** So most do not persist long enough to reach the stratosphere where UV would release their chlorine **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-112; AQA spec 3.2.1.3.	
08.1	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Volume of ash is much less than the original waste (less landfill) **1** Heat used for district heating / electricity generation (energy recovery) **1** No sorting / complicated management needed **1** Destroys pathogens / flammable organic wastes converted to CO2 and water **1** Refuse-derived fuel can replace fossil fuels (eg cement kilns) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3

Do not write outside the box

Question	Marking guidance	Marks
08.2	**1 mark each** 1 mark for each of the 4 points below. **1** High temperature and rapid cooling reduce the formation of toxic dioxins (from organic wastes reacting with chlorine) **1** Scrubber dissolves hydrogen chloride (from chlorine in PVC) which is then neutralised with alkali / limestone **OR** removes heavy metals for storage at high pH **1** Wet waste needs extra fuel to burn and lowers combustion temperature – composting recovers its nutrient value **1** Recycling conserves the resource value of metals, glass and plastics that would otherwise be lost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Recycling** → reprocessing a material to make something new **Do not credit** • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Ash may contain concentrated heavy metals / toxic residues **1** Must be stored in solid form at high pH / hazardous waste landfill so the metals remain insoluble **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
08.4	**Any 1 of 2** 1 mark each for any 1 of the 2 points below, in any order, maximum 1. **1** Fuel (natural gas) needed to burn wet / non-flammable wastes **1** Cost of treating effluent gases (SO2, NOx, HCl, smoke) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 283-284; AQA spec 3.4.3.2.12.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** (150 - 125) / 150 x 100 = 16.7% (accept 17%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** A fixed fee gives no incentive to save water because the bill is the same however much is used **1** Metering matches the charge to the volume used so households save money by using less (fewer hosepipes, low-flow appliances) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1 Low-water appliances:** dual-flush toilets, low-flow shower heads, efficient washing machines / dishwashers, tap timers **1 Greywater re-use:** bath / shower water used to flush toilets or water gardens **1** Rainwater collection / water butts for garden use **1 Xeriscaping:** planting drought-adapted species that need no watering **1** Repairing leaks / dripping taps **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** Old infrastructure / pipes in poor condition leak **1** Repair and good maintenance / pipe replacement / leak detection reduce wastage **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
09.5	**1 mark each** 1 mark for each of the 2 points below. **1** Water is delivered directly to the plant roots / soil around each plant **1** Less evaporation loss from spray droplets and from wetting the whole soil surface – less runoff **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 124-126; AQA spec 3.2.2.5.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **State of matter:** solids deposit near the source, gases disperse widely in the atmosphere – density (lead dust, hydrogen cyanide at Bhopal) controls how far particles/gases travel • **Persistence / half-life:** persistent pollutants (organochlorines, CFCs, heavy metals, radioactive wastes with long half-lives) travel further, accumulate and need containment rather than dilution – non-persistent pollutants (sewage, organophosphates) can be allowed to degrade • Toxicity, specificity, mode of action (enzyme inhibition, neurotoxins) determine acceptable concentrations and which species are at risk – LD50 / MDAF / MDNF used to set doses • Solubility in water (nitrates disperse and leach - control by nitrate vulnerable zones) vs liposolubility (DDT, mercury bioaccumulate and biomagnify - control by bans / restrictions) • **Reactivity:** reactive pollutants may degrade quickly or form secondary pollutants (ozone, acid rain, PANs) - controlling primary pollutants (catalytic converters, FGD) prevents secondary pollutants • Adsorption onto clay / sediments immobilises metals and caesium – can be used deliberately (activated carbon filters, polymer adsorption) • Mutagenic / carcinogenic / teratogenic action means no safe dose – exposure kept As Low As Reasonably Achievable (ALARA) • Synergism (ozone + SO2; smoke + SO2) means pollutants must be considered together • Environmental features (temperature inversions, currents, pH, oxygen) interact with properties: Critical Pathway Analysis uses properties + environment to predict where pollutants go and to focus monitoring • **Evaluation:** properties of new chemicals can be predicted from similar ones (precautionary principle)	9

Do not write outside the box

Question	Marking guidance	Marks
	– understanding allows prevention rather than clean-up **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **Persistence** → does not break down / stays in the environment for a long time • **half-life** → the time for the activity, or the amount, to fall by half • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	
	Model answer - Level 3 (9 marks)	
	A pollutant's properties largely determine where it goes and how much harm it does, so they can be used both to predict its behaviour and to choose a control. State of matter and density govern dispersal: solids deposit near the source, gases spread widely, and dense gases such as hydrogen cyanide at Bhopal hug the ground. Persistence decides whether dilution is enough: non-persistent wastes like sewage can be discharged where they break down, but organochlorines, CFCs, heavy metals and long-lived radioisotopes travel far and accumulate, so they need containment, bans or restrictions instead. Toxicity, specificity and mode of action, quantified by LD50 and used to set acceptable doses, identify which species are at risk. Solubility explains why nitrate leaches and is controlled by nitrate vulnerable zones, while liposolubility explains why DDT and methyl mercury bioaccumulate and biomagnify along food chains. Reactive pollutants may degrade quickly or form secondary pollutants - ozone, acid rain, PANs - so controlling the primary emissions with catalytic converters or flue-gas desulfurisation prevents the secondary ones. Adsorption on clay immobilises metals and caesium and is exploited deliberately in activated-carbon and polymer filters. Mutagens and carcinogens have no safe dose, so exposure is kept As Low As Reasonably Achievable, and synergism between ozone and sulfur dioxide, or smoke and sulfur dioxide, means pollutants must be assessed together.	
	Properties interact with the environment - inversions, currents, pH, oxygen - which is why Critical Pathway Analysis combines both to predict where pollutants concentrate and to focus monitoring. The greatest value of this understanding is preventive: the properties of a new chemical can be predicted from similar ones under the precautionary principle, allowing control before release rather than clean-up afterwards.	
	Weaker answer - Level 1 (2 marks)	
	Pollutants have properties like toxicity, persistence and solubility. Toxic pollutants are dangerous. Persistent pollutants like DDT last a long time in the environment and build up in food chains. Soluble pollutants dissolve in water and get into rivers. Knowing these properties helps scientists to predict what pollutants will do and to control them, for example by banning DDT.	
	What makes the difference	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 1 defines three properties in isolation; Level 3 pairs every property with a prediction *and* a control strategy, which is what the question asks.	
	• Level 3 uses a wide range of accurate examples (Bhopal, nitrate zones, methyl mercury, PANs, caesium adsorption); Level 1 uses DDT twice.	
	• The final paragraph links properties to environmental conditions (CPA) and to the precautionary principle - synoptic evaluation that distinguishes the top band.	
10.2	1 mark for the point below. **1** The increase in the amount / concentration of a substance within an (individual) organism over time, because intake exceeds excretion or breakdown **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **bioaccumulation** → a pollutant building up inside one organism over time **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240; AQA spec 3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Features of fertile soil:** water, nutrients, air, organic matter, pH, biota, texture, structure, depth – slow rate of soil formation vs rapid erosion (dynamic equilibrium broken) • **Causes of degradation:** vegetation removal, ploughing vulnerable soils, overgrazing, reduced organic matter and biota, cultivating steep slopes, compaction, salinisation from irrigation, nutrient depletion, pollution • **Types of erosion:** rain splash, runoff, wind, slumping – consequences: reduced productivity, river and reservoir sedimentation, flooding, coastal sedimentation, particulates, desertification, landslides • USLE as a tool for estimating loss and testing management • **Management:** long-term crops, low / zero tillage, contour ploughing, tied ridging, terracing, stone rows, windbreaks, multicropping / strip cropping, mulching, organic matter, livestock control, drainage / pH / nutrient management, conservation of soil biota • Links to sustainable agriculture, carbon storage, hydrological cycle, circular economy	25
	• Evaluation and judgement on which strategies are most effective in different environments *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169, 328-337; AQA spec 3.2.5, 3.5.1.3, 3.5.1.5.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Negative feedback:** increased low-level cloud (albedo), increased photosynthesis and sequestration – hydrological cycle evaporation-precipitation • **Positive feedback:** melting permafrost (methane), reduced albedo of snow / ice / sea ice, methane hydrate release, ocean acidification and coral loss, forest and peat fires, soil decomposition, water vapour, cirrus clouds • **Tipping points:** self-sustaining processes – why control must precede them – examples • **Implications:** urgency and precautionary principle – need for deep cuts in greenhouse gases (energy, transport, buildings, agriculture, forestry, CCS) rather than only adaptation – geoengineering risks • Difficulty of prediction because feedbacks interact at different scales – models – international agreements and per capita footprints • Judgement on whether current strategies are adequate	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-107, 371-372; AQA spec 3.2.1.2, 3.6.1.	

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